

Class 14: RNASeq Mini-Project

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##Background

Our Data for today comes from a HOX gene knock-out study

Trapnell C, Hendrickson DG, Sauvageau M, Goff L et al. “Differential analysis of gene regulation at transcript resolution with RNA-seq”. Nat Biotechnol 2013 Jan;31(1):46-53. PMID: 23222703

The authors report on differential analysis of lung fibroblasts in response to loss of the developmental transcription factor HOXA1.

Data Import

We have 2 keyinput files: counts and metadata.

```
metaFile <- "data/GSE37704_metadata.csv"
countFile <- "data/GSE37704_featurecounts.csv"

# Import metadata and take a peek
colData = read.csv("GSE37704_metadata.csv", row.names=1)
head(colData)
```

```
      condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
# Import countdata
countData = read.csv("GSE37704_featurecounts.csv", row.names=1)
head(countData)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28
ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0
ENSG00000187634	3214	124	123	205	207	212
	SRR493371					
ENSG00000186092	0					
ENSG00000279928	0					
ENSG00000279457	46					
ENSG00000278566	0					
ENSG00000273547	0					
ENSG00000187634	258					

Q. Complete the code below to remove the troublesome first column from countData

We need to remove the first length column from countdata to have a 1:1 correspondents with colData rows.

```
countData <- countData[,-1]
```

```
rownames(colData) == colnames(countData)
```

```
[1] TRUE TRUE TRUE TRUE TRUE TRUE
```

Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

Tip: What will rowSums() of countData return and how could you use it in this context?

```
# Filter count data where you have 0 read count across all samples.  
# countData = countData[rowSums(countData) > 0,]  
# head(countData)
```

Check and Tidy

```
library(DESeq2)
```

Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following objects are masked from 'package:base':

```
as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,  
setequal, union
```

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
unsplit, which.max, which.min

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

```
colAlls, colAnyNAs, colAnys, colAvgPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars
```

Loading required package: Biobase

Welcome to Bioconductor

```
Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.
```

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

```
rowMedians
```

The following objects are masked from 'package:matrixStats':

```
anyMissing, rowMedians
```

Setup for DESeq

```
dds <- DESeqDataSetFromMatrix(countData = countData,
                              colData = colData,
                              design = ~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

Run DESeq

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

Get results

```
res <- results(dds)
head(res)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000186092	0.0000	NA	NA	NA	NA
ENSG00000279928	0.0000	NA	NA	NA	NA

ENSG00000279457	29.9136	0.179257	0.324822	0.551863	0.58104205
ENSG00000278566	0.0000	NA	NA	NA	NA
ENSG00000273547	0.0000	NA	NA	NA	NA
ENSG00000187634	183.2296	0.426457	0.140266	3.040350	0.00236304
	padj				
	<numeric>				
ENSG00000186092	NA				
ENSG00000279928	NA				
ENSG00000279457	0.68707978				
ENSG00000278566	NA				
ENSG00000273547	NA				
ENSG00000187634	0.00516278				

Q. Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```
summary(res)
```

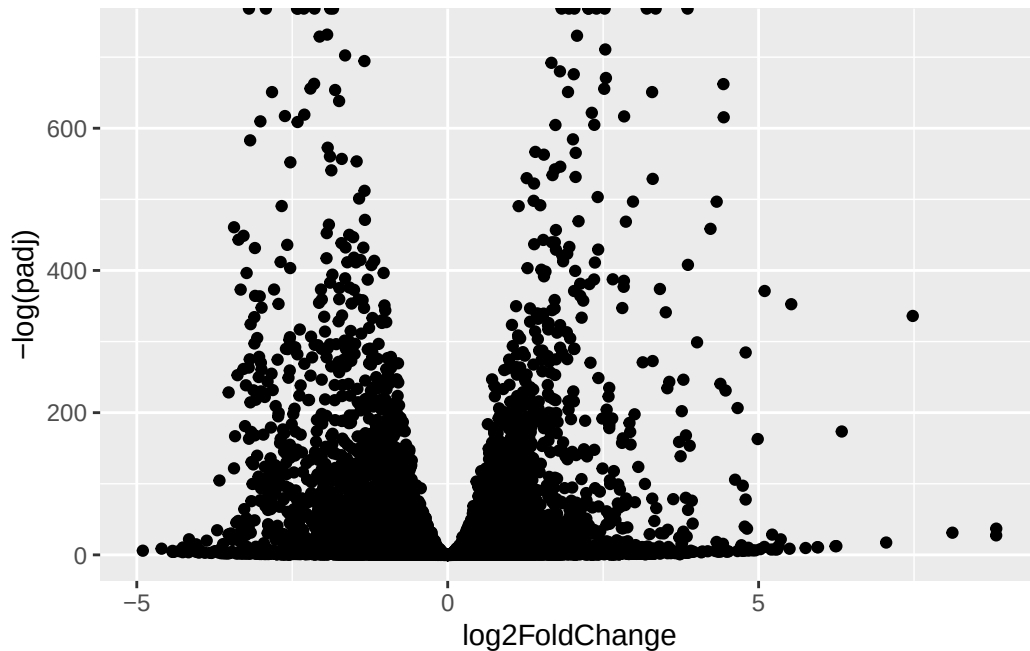
```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4393, 27%
outliers [1]      : 0, 0%
low counts [2]    : 1221, 7.6%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Volcano plot

```
library(ggplot2)

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point()
```

Warning: Removed 5054 rows containing missing values or values outside the scale range (``geom_point()``).

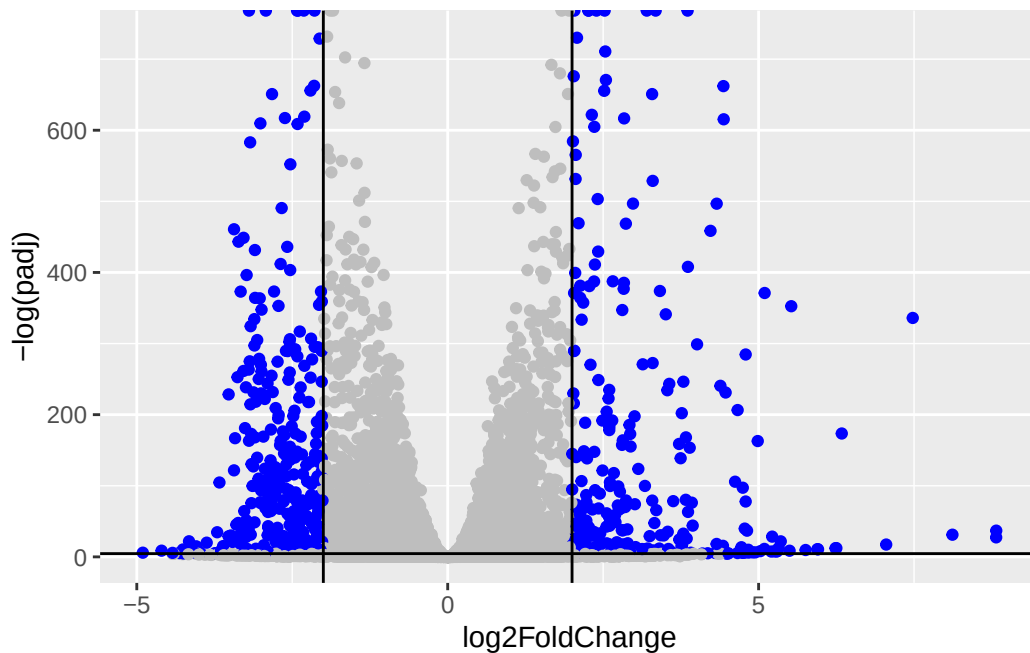


Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

```
mycols <- rep("gray", nrow(res))
mycols[abs(res$log2FoldChange) > 2] <- "blue"
mycols[res$padj > 0.01] <- "gray"
```

```
ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point(col = mycols) +
  geom_vline(xintercept = c(-2,2)) +
  geom_hline(yintercept = -log(0.01))
```

Warning: Removed 5054 rows containing missing values or values outside the scale range (`geom\_point()`).



Add anotation

Q. Use the `mapIDs()` function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

```
library("AnnotationDbi")
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

[1] "ACCNUM"	"ALIAS"	"ENSEMBL"	"ENSEMBLPROT"	"ENSEMBLTRANS"
[6] "ENTREZID"	"ENZYME"	"EVIDENCE"	"EVIDENCEALL"	"GENENAME"
[11] "GENETYPE"	"GO"	"GOALL"	"IPI"	"MAP"
[16] "OMIM"	"ONTOLOGY"	"ONTOLOGYALL"	"PATH"	"PFAM"
[21] "PMID"	"PROSITE"	"REFSEQ"	"SYMBOL"	"UCSCKG"
[26] "UNIPROT"				

```
res$symbol = mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="SYMBOL",
                    multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez = mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="ENTREZID",
                    multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$name = mapIds(org.Hs.eg.db,
                  keys=row.names(res),
                  keytype="ENSEMBL",
                  column="GENENAME",
                  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res, 10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000186092	0.0000	NA	NA	NA	NA
ENSG00000279928	0.0000	NA	NA	NA	NA
ENSG00000279457	29.9136	0.1792571	0.3248216	0.551863	5.81042e-01
ENSG00000278566	0.0000	NA	NA	NA	NA
ENSG00000273547	0.0000	NA	NA	NA	NA
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630158	1.43990e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08

ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.9798	0.5428105	0.5215598	1.040744	2.97994e-01
	padj	symbol	entrez		name
	<numeric>	<character>	<character>		<character>
ENSG00000186092	NA	OR4F5	79501	olfactory receptor f..	
ENSG00000279928	NA	NA	NA		NA
ENSG00000279457	6.87080e-01	NA	NA		NA
ENSG00000278566	NA	NA	NA		NA
ENSG00000273547	NA	NA	NA		NA
ENSG00000187634	5.16278e-03	SAMD11	148398	sterile alpha motif ..	
ENSG00000188976	1.76741e-35	NOC2L	26155	NOC2 like nucleolar ..	
ENSG00000187961	1.13536e-07	KLHL17	339451	kelch like family me..	
ENSG00000187583	9.18988e-01	PLEKHN1	84069	pleckstrin homology ..	
ENSG00000187642	4.03817e-01	PERM1	84808	PPARGC1 and ESRR ind..	

Save annotated results

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
res = res[order(res$pvalue),]
write.csv(res, file="results_annotated.csv")
```

MapIDs

```
library(pathview)
```

```
#####
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.
```

```
The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG
license agreement (details at http://www.kegg.jp/kegg/legal.html).
```

```
#####
```

```
library(gage)
```

```
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

```
$`hsa00230 Purine metabolism`
```

```
[1] "100" "10201" "10606" "10621" "10622" "10623" "107" "10714"
[9] "108" "10846" "109" "111" "11128" "11164" "112" "113"
[17] "114" "115" "122481" "122622" "124583" "132" "158" "159"
[25] "1633" "171568" "1716" "196883" "203" "204" "205" "221823"
[33] "2272" "22978" "23649" "246721" "25885" "2618" "26289" "270"
[41] "271" "27115" "272" "2766" "2977" "2982" "2983" "2984"
[49] "2986" "2987" "29922" "3000" "30833" "30834" "318" "3251"
[57] "353" "3614" "3615" "3704" "377841" "471" "4830" "4831"
[65] "4832" "4833" "4860" "4881" "4882" "4907" "50484" "50940"
[73] "51082" "51251" "51292" "5136" "5137" "5138" "5139" "5140"
[81] "5141" "5142" "5143" "5144" "5145" "5146" "5147" "5148"
[89] "5149" "5150" "5151" "5152" "5153" "5158" "5167" "5169"
[97] "51728" "5198" "5236" "5313" "5315" "53343" "54107" "5422"
```

```
[105] "5424" "5425" "5426" "5427" "5430" "5431" "5432" "5433"
[113] "5434" "5435" "5436" "5437" "5438" "5439" "5440" "5441"
[121] "5471" "548644" "55276" "5557" "5558" "55703" "55811" "55821"
[129] "5631" "5634" "56655" "56953" "56985" "57804" "58497" "6240"
[137] "6241" "64425" "646625" "654364" "661" "7498" "8382" "84172"
[145] "84265" "84284" "84618" "8622" "8654" "87178" "8833" "9060"
[153] "9061" "93034" "953" "9533" "954" "955" "956" "957"
[161] "9583" "9615"
```

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
      1266      54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288
```

```
# Get the results
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```
attributes(keggres)
```

```
$names
[1] "greater" "less" "stats"
```

```
# Look at the first few down (less) pathways
head(keggres$less)
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	7.077982e-06	-4.432593	7.077982e-06
hsa03030 DNA replication	9.424076e-05	-3.951803	9.424076e-05
hsa03013 RNA transport	1.160132e-03	-3.080629	1.160132e-03
hsa04114 Oocyte meiosis	2.563806e-03	-2.827297	2.563806e-03
hsa03440 Homologous recombination	3.066756e-03	-2.852899	3.066756e-03
hsa00010 Glycolysis / Gluconeogenesis	4.360092e-03	-2.663825	4.360092e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001160789	124	7.077982e-06
hsa03030 DNA replication	0.007727742	36	9.424076e-05
hsa03013 RNA transport	0.063420543	149	1.160132e-03
hsa04114 Oocyte meiosis	0.100589607	112	2.563806e-03
hsa03440 Homologous recombination	0.100589607	28	3.066756e-03
hsa00010 Glycolysis / Gluconeogenesis	0.119175854	65	4.360092e-03

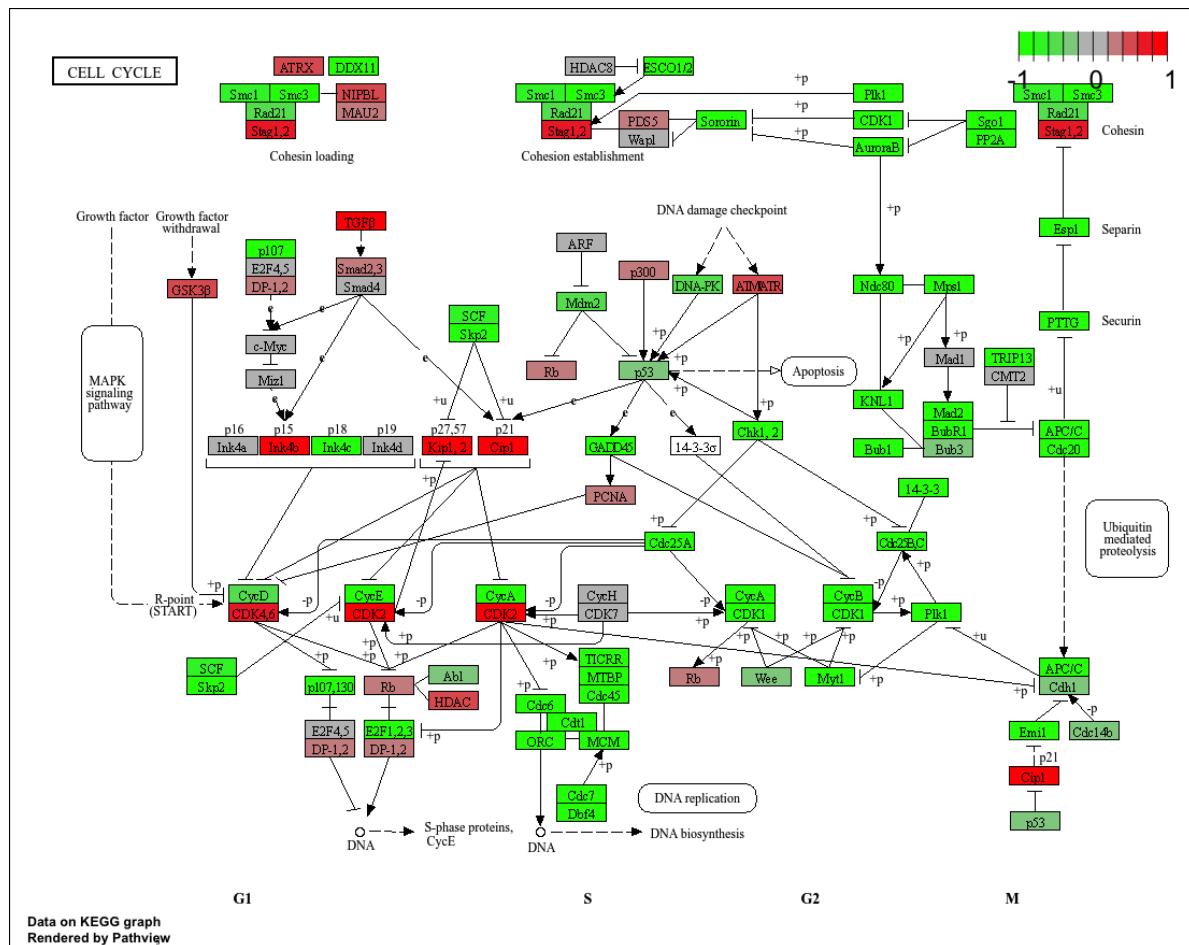
Pathway

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/nhidinh/Desktop/BIMM143/Class 14

Info: Writing image file hsa04110.pathview.png



```
# A different PDF based output of the same data
```

```
pathview(gene.data=foldchanges, pathway.id="hsa04110", kegg.native=FALSE)
```

'select()' returned 1:1 mapping between keys and columns

Warning: reconcile groups sharing member nodes!

```
      [,1] [,2]  
[1,] "9"  "300"  
[2,] "9"  "306"
```

Info: Working in directory /Users/nhidinh/Desktop/BIMM143/Class 14

Info: Writing image file hsa04110.pathview.pdf

```
## Focus on top 5 upregulated pathways here for demo purposes only  
keggrespathways <- rownames(keggres$greater)[1:5]  
  
# Extract the 8 character long IDs part of each string  
keggresids = substr(keggrespathways, start=1, stop=8)  
keggresids
```

```
[1] "hsa04740" "hsa04640" "hsa00140" "hsa04630" "hsa04976"
```

```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/nhidinh/Desktop/BIMM143/Class 14

Info: Writing image file hsa04740.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/nhidinh/Desktop/BIMM143/Class 14

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/nhidinh/Desktop/BIMM143/Class 14

Info: Writing image file hsa00140.pathview.png

'select()' returned 1:1 mapping between keys and columns

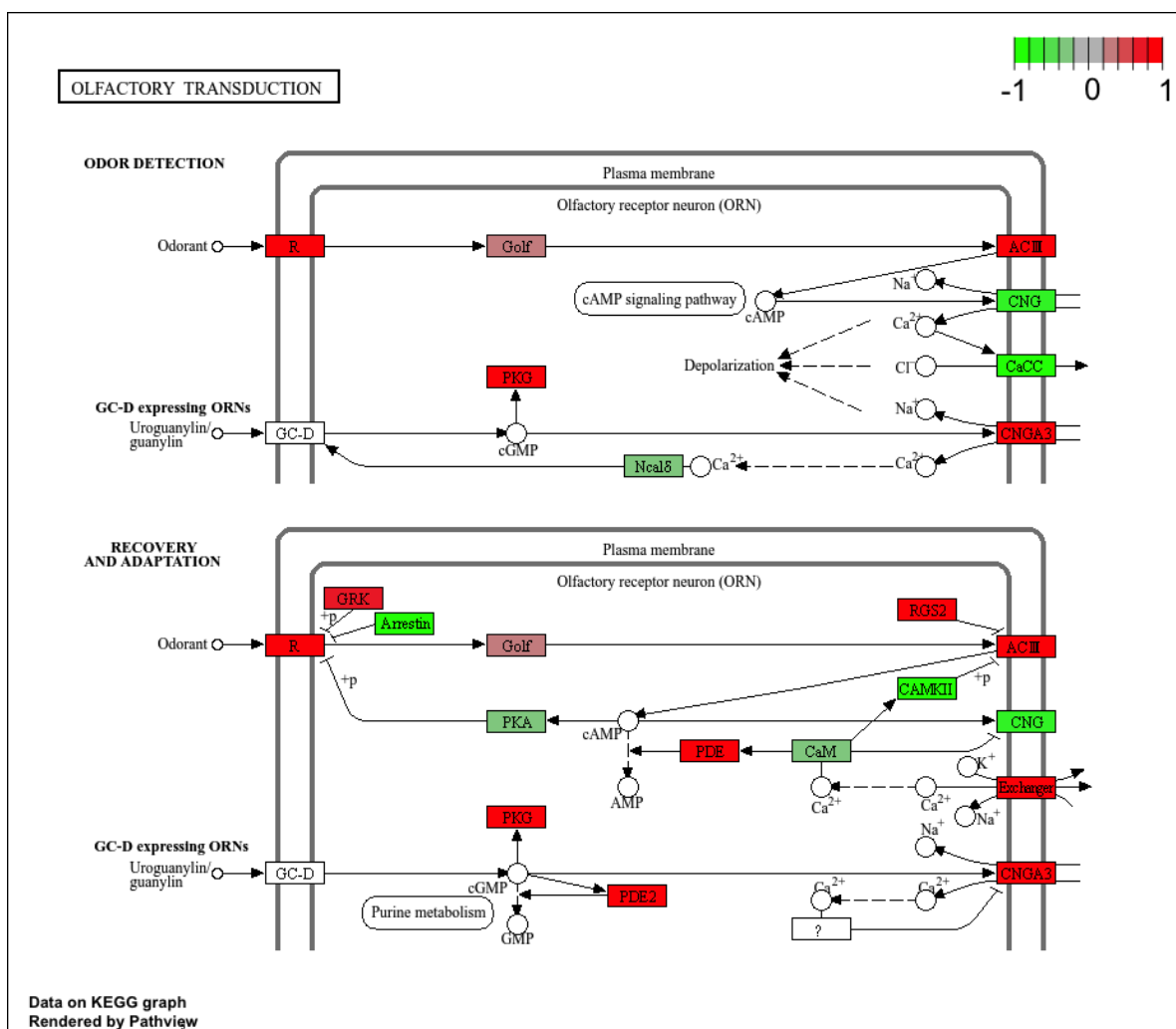
Info: Working in directory /Users/nhidinh/Desktop/BIMM143/Class 14

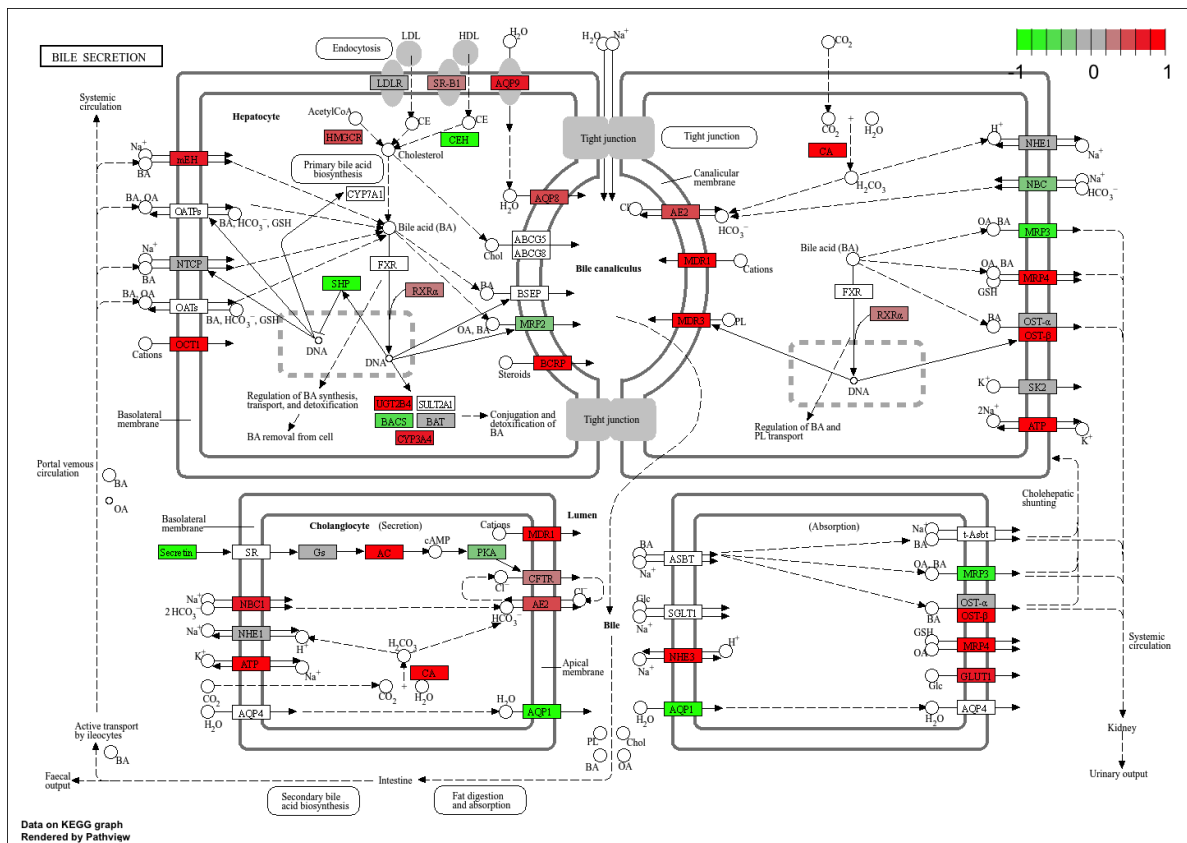
Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/nhidinh/Desktop/BIMM143/Class 14

Info: Writing image file hsa04976.pathview.png





Q. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
## Focus on top 5 upregulated pathways here for demo purposes only
keggrespathways <- rownames(keggres$less)[1:5]
```

```
# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa04114" "hsa03440"
```

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/nhidinh/Desktop/BIMM143/Class 14

Info: Writing image file hsa04110.pathview.png

```
pathview(gene.data=foldchanges, pathway.id="hsa03030")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/nhidinh/Desktop/BIMM143/Class 14

Info: Writing image file hsa03030.pathview.png

```
pathview(gene.data=foldchanges, pathway.id="hsa03013")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/nhidinh/Desktop/BIMM143/Class 14

Info: Writing image file hsa03013.pathview.png

```
pathview(gene.data=foldchanges, pathway.id="hsa04114")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/nhidinh/Desktop/BIMM143/Class 14

Info: Writing image file hsa04114.pathview.png

```
pathview(gene.data=foldchanges, pathway.id="hsa03440")
```

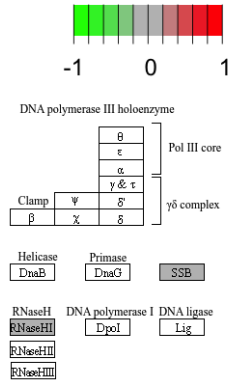
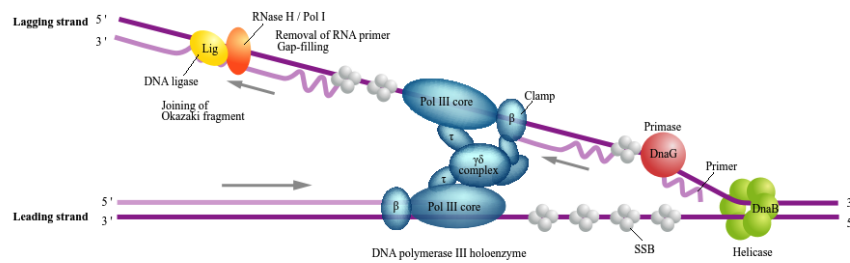
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/nhidinh/Desktop/BIMM143/Class 14

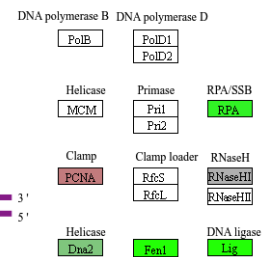
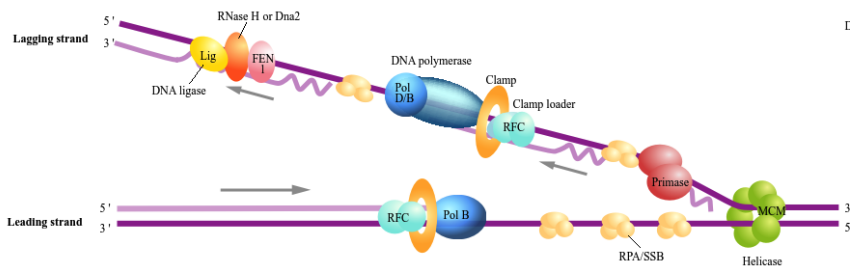
Info: Writing image file hsa03440.pathview.png

DNA REPLICATION

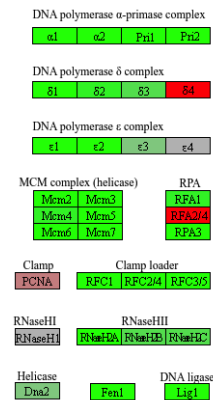
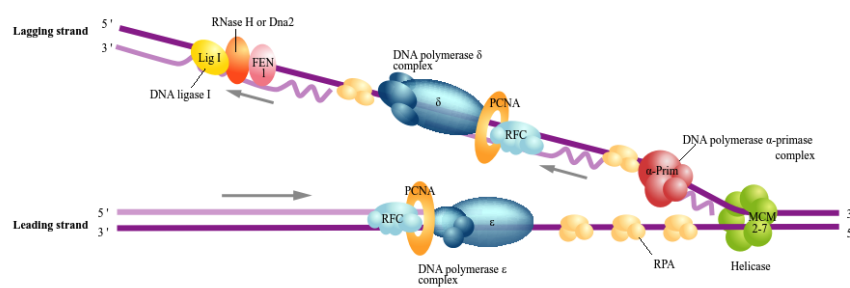
Replication complex (Bacteria)



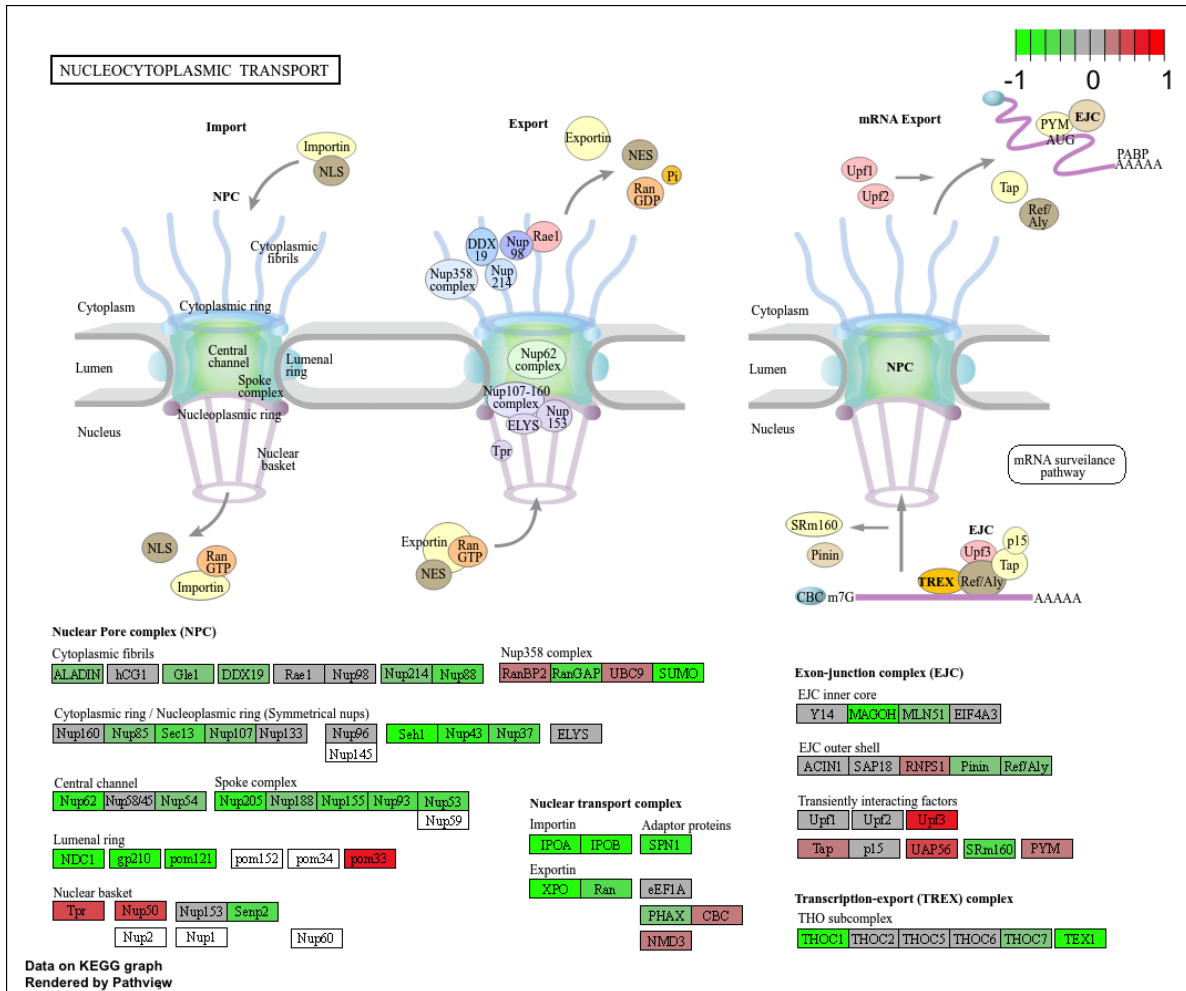
Replication complex (Archaea)



Replication complex (Eukaryotes)



Data on KEGG graph
Rendered by Pathview



	p.geomean	stat.mean	p.val
G0:0007156 homophilic cell adhesion	1.734864e-05	4.210777	1.734864e-05
G0:0048729 tissue morphogenesis	5.407952e-05	3.888470	5.407952e-05
G0:0002009 morphogenesis of an epithelium	5.727599e-05	3.878706	5.727599e-05
G0:0030855 epithelial cell differentiation	2.053700e-04	3.554776	2.053700e-04
G0:0060562 epithelial tube morphogenesis	2.927804e-04	3.458463	2.927804e-04
G0:0048598 embryonic morphogenesis	2.959270e-04	3.446527	2.959270e-04

	q.val	set.size	expl
G0:0007156 homophilic cell adhesion	0.07584825	137	1.734864e-05
G0:0048729 tissue morphogenesis	0.08347021	483	5.407952e-05
G0:0002009 morphogenesis of an epithelium	0.08347021	382	5.727599e-05
G0:0030855 epithelial cell differentiation	0.16449701	299	2.053700e-04
G0:0060562 epithelial tube morphogenesis	0.16449701	289	2.927804e-04
G0:0048598 embryonic morphogenesis	0.16449701	498	2.959270e-04

\$less

	p.geomean	stat.mean	p.val
G0:0048285 organelle fission	6.626774e-16	-8.170439	6.626774e-16
G0:0000280 nuclear division	1.797050e-15	-8.051200	1.797050e-15
G0:0007067 mitosis	1.797050e-15	-8.051200	1.797050e-15
G0:0000087 M phase of mitotic cell cycle	4.757263e-15	-7.915080	4.757263e-15
G0:0007059 chromosome segregation	1.081862e-11	-6.974546	1.081862e-11
G0:0051301 cell division	8.718528e-11	-6.455491	8.718528e-11

	q.val	set.size	expl
G0:0048285 organelle fission	2.618901e-12	386	6.626774e-16
G0:0000280 nuclear division	2.618901e-12	362	1.797050e-15
G0:0007067 mitosis	2.618901e-12	362	1.797050e-15
G0:0000087 M phase of mitotic cell cycle	5.199689e-12	373	4.757263e-15
G0:0007059 chromosome segregation	9.459800e-09	146	1.081862e-11
G0:0051301 cell division	6.352901e-08	479	8.718528e-11

\$stats

	stat.mean	expl
G0:0007156 homophilic cell adhesion	4.210777	4.210777
G0:0048729 tissue morphogenesis	3.888470	3.888470
G0:0002009 morphogenesis of an epithelium	3.878706	3.878706
G0:0030855 epithelial cell differentiation	3.554776	3.554776

```
G0:0060562 epithelial tube morphogenesis      3.458463 3.458463
G0:0048598 embryonic morphogenesis            3.446527 3.446527
```

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8146"
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

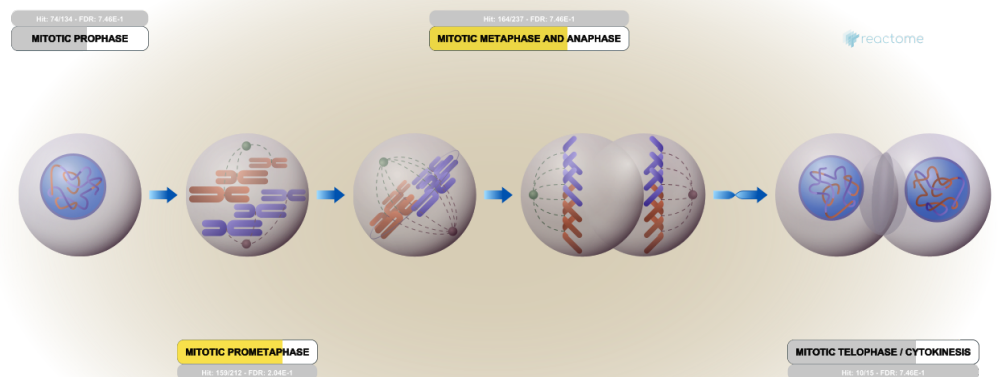


Figure from Reactome:

> Q. Q: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The most significant pathway by “Entities p-value” in Reactome is typically Cell Cycle, Mitotic or DNA Replication. These results strongly match previous KEGG findings, which also highlighted “Cell cycle” and “DNA replication” as the top down-regulated pathways. The minor differences in rankings or p-values between the two methods are primarily caused by differences in how each database is curated, the specific hierarchical structure of their pathway maps, and the unique gene-to-pathway assignment criteria used by Reactome compared to the KEGG database.